Supplementary Appendix

Supplementary Appendix A. Graph-learning and target-extraction notation

Table 1: GNN notation.

Symbol	Type	Definition
I	Set	Zones (graph nodes).
E	Set	Graph edges between zones.
T	Set	Time steps used for learning.
P	Parameter	Number of time-varying mobility features.
R	Parameter	Number of static zone attributes.
L	Parameter	Number of spatial message-passing layers.
$\mathcal{G}$	Object	Weighted urban graph $(I, E, \mathbf{W})$.
$\mathcal{N}(i)$	Set	Neighbor set of zone i.
w_{ij}	Parameter	Edge weight from zone i to zone j.
t_{ij}	Parameter	Network travel time between zones.
τ	Parameter	Scale parameter in edge-weight decay.
$\mathbf{x}_{i,t}$	Input vector	Node features for zone i at time t.
$\mathbf{X}_t$	Input matrix	Stacked node features at time t.
$\mathbf{W}_s^{(l)}, \mathbf{U}_s^{(l)}$	Parameter	Trainable spatial weight matrices at layer l.
$\sigma(\cdot)$	Operator	Nonlinear activation in the spatial propagation step.
$\mathbf{h}_{i,t}^{(l)}$	Latent state	Spatial embedding at layer l.
$\mathbf{g}_{i,t}$	Latent state	Temporal embedding after recurrent update.
$f_{\text{out}}(\cdot)$	Operator	Output head mapping latent states to predicted demand.
$\hat{y}_{i,t}$	Output	Predicted charging-demand intensity.
$y_{i,t}$	Target	Observed/proxy demand intensity for training.
T_{train}	Set	Training subset of time steps.
$\mathbf{r}_t$	Latent state	Mean latent representation used in temporal attention.
$\mathbf{q}, \mathbf{W}_a$	Parameter	Trainable attention parameters.
α_t	Output	Temporal attention weight at time t.
s_i^{raw}	Derived output	Raw zone-level priority score from ST-GNN outputs.
s_i	Derived output	Normalized priority index used in optimization.
T_v	Set	Venue-relevant time window associated with venue type v.
ω_{iv}	Derived output	Venue-specific share weight for zone i.

Supplementary Appendix B. Optimization-and-equity notation

Table 2: Optimization and equity notation.

Symbol	Type	Definition
V	Set	Venue types (residential, workplace, public).
M	Set	Vulnerability indicators.
$\mathcal{V}^Q$	Set	Most vulnerable quartile of zones used for quartile-share and coverage reporting.
$\mathcal{V}^H$	Set	High-vulnerability zones for floor constraints.
Q_v	Parameter	Exogenous quota of ports for venue v.
π_{iv}	Parameter	Share for zone i, venue v, from the target-allocation equation in the main text.
T_{iv}	Parameter	Priority-preserving target ports for zone i, venue v.
z_{im}	Parameter	Standardized vulnerability indicator m in zone i.
η_m	Parameter	Indicator weight in vulnerability index.
ν_i	Parameter	Social charging vulnerability index for zone i.
λ_v	Parameter	Venue-specific decay coefficient in accessibility (minutes).
ε	Parameter	Tolerance for loss of priority fidelity.
$U^\star$	Parameter	Optimal upper-level objective value.
$\underline{A}$	Parameter	Accessibility floor for $\mathcal{V}^H$.
κ	Parameter	Intensity parameter for vulnerability weighting in the socially weighted equity objective.
S^Q	Metric	Share of total accessibility captured by $\mathcal{V}^Q$.
C^{15}	Metric	Share of zones in $\mathcal{V}^Q$ with at least one allocated port reachable within 15 minutes.
G_{W}	Metric	Vulnerability-weighted accessibility Gini computed with zone weights $1+\nu_i$.
$\bar{\nu}_{ij}$	Parameter	Pairwise vulnerability weight used in the socially weighted lower-level objective.
γ	Parameter	Relaxation factor used to define the soft accessibility floor in S3.
$q_{0.10}(\cdot)$	Operator	Tenth-percentile operator used to calibrate the soft floor.
β	Parameter	Penalty weight on protected-zone shortfalls in S3.
n_{iv}	Decision variable	Integer ports allocated to zone i, venue v.
u_{iv}	Decision variable	Absolute deviation from target T_{iv}.
A_i	Metric	Accessibility experienced by demand zone i.
z_{ij}	Decision variable	Linearized absolute accessibility gap between i and j.
d_i	Decision variable	Nonnegative shortfall variable for the soft floor in protected zones.

Supplementary Appendix C. Additional figures

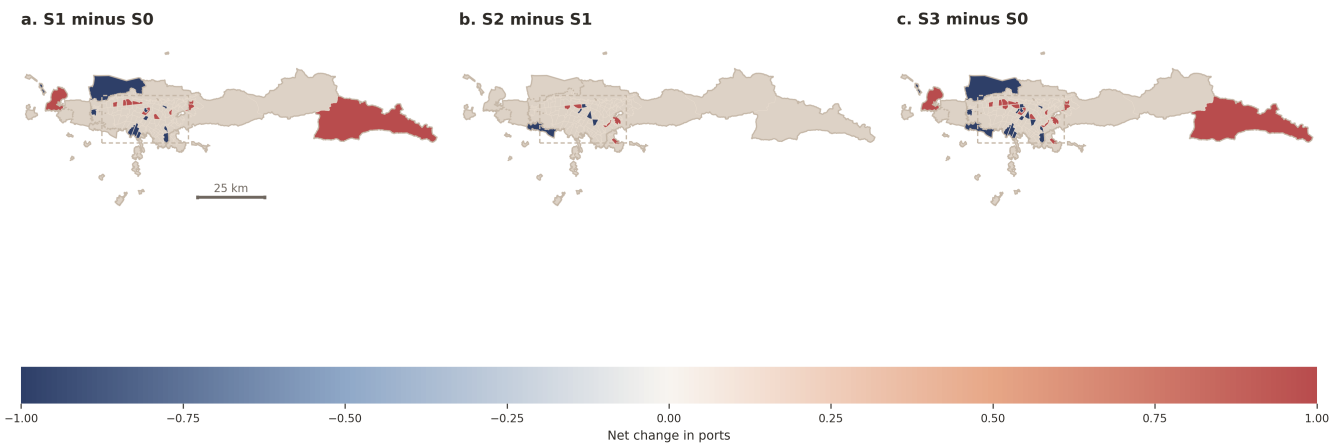


Figure 1: Growth-progression maps across allocation stages. The panels expand the stage-to-stage spatial story behind the final allocation comparison and provide supplementary spatial detail on how the scenario sequence evolves under fixed quotas.

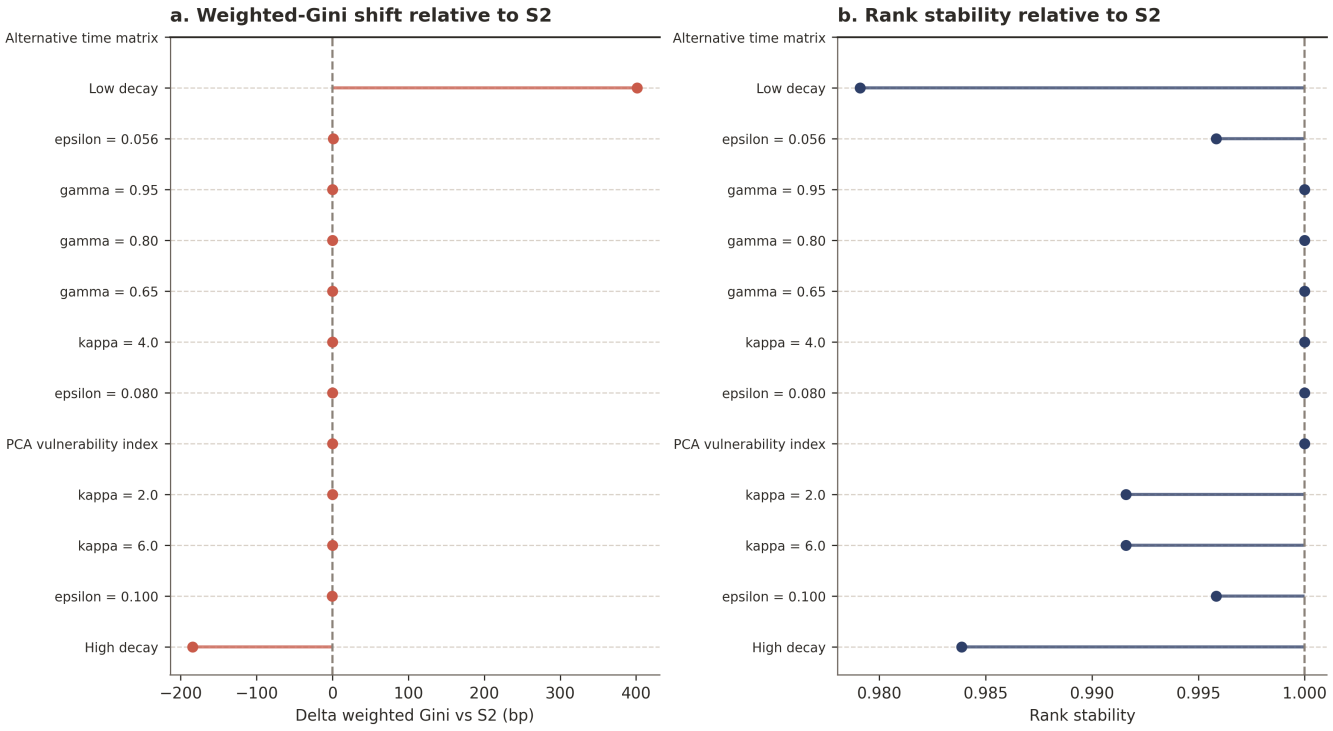


Figure 2: Robustness checks around the S2 baseline. The main scenario ordering and high rank stability are preserved across alternative decay, vulnerability-index, epsilon, kappa, and soft-floor specifications, while the strongest sensitivity appears in the accessibility-decay calibration.